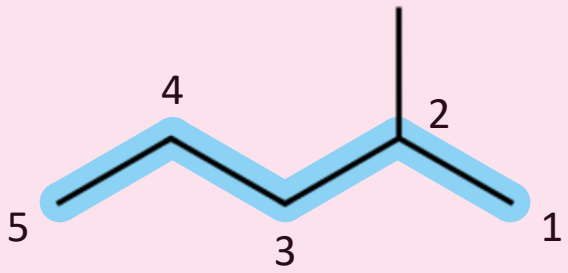
A01 FOUNDATION Part A · Structure → Name

Molecular formula: C_6H_{14}

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 5 C → pentane.

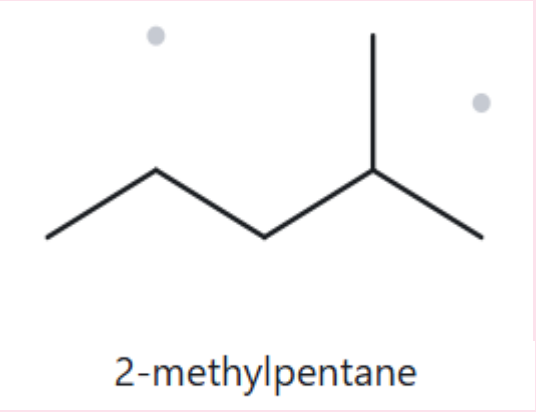
Numbering: left → right gives 4; right → left gives 2. 2 < 4, so number from the right.

Substituents: methyl on C2 → 2-methyl.

3 · MY NAME

2-methylpentane

4 · THE ALKANE CHECK



Returned name:

2-methylpentane

✓ **Match**

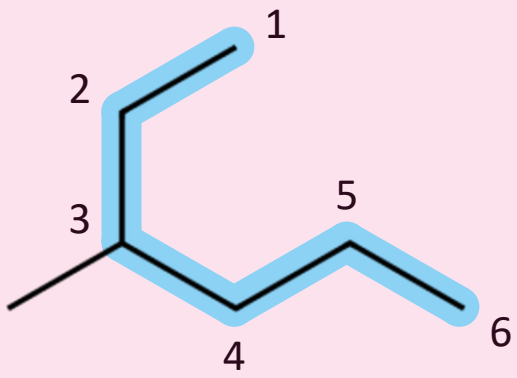
Same as my name (2-methylpentane): parent, locants, substituents and punctuation all agree.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A02 FOUNDATION Part A · Structure → Name

Molecular formula: C_7H_{16}

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 6 C → hexane (not the horizontal line: top end → branch carbon → lower zigzag).

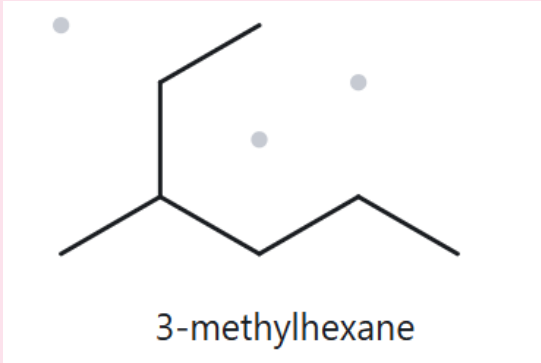
Numbering: top end → bottom-right end gives 3; bottom-right end → top end gives 4. 3 < 4, so number from the top end.

Substituents: methyl on C3 → 3-methyl.

3 · MY NAME

3-methylhexane

4 · THE ALKANE CHECK



Returned name:

3-methylhexane

✓ **Match**

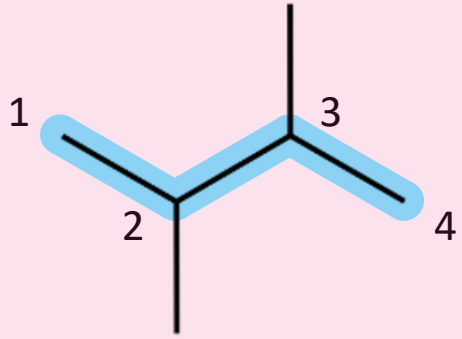
Same as my name (3-methylhexane): parent, locants, substituents and punctuation all agree.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A03 FOUNDATION Part A · Structure → Name

Molecular formula: C_6H_{14}

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 4 C → butane.

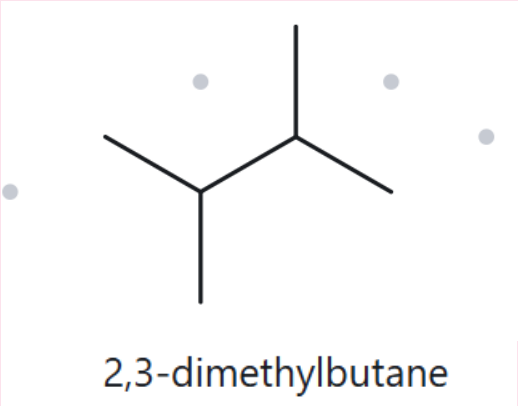
Numbering: left → right gives 2,3; right → left gives 2,3. Tie: both directions give the same name.

Substituents: methyl on C2 and on C3 → 2,3-dimethyl.

3 · MY NAME

2,3-dimethylbutane

4 · THE ALKANE CHECK



Returned name:

2,3-dimethylbutane

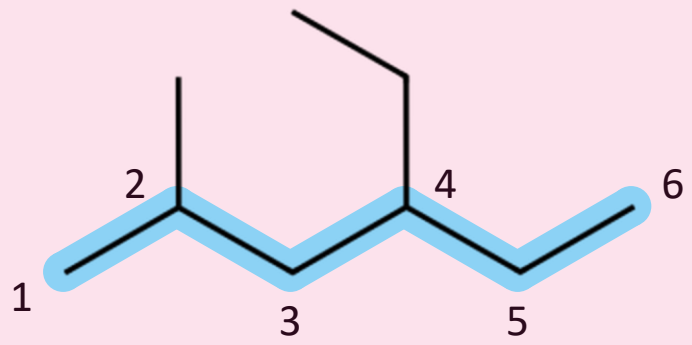
✓ **Match**

Same as my name (2,3-dimethylbutane): parent, locants, substituents and punctuation all agree.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

Molecular formula: C_9H_{20}

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 6 C → hexane (the equal-length chains all carry one methyl and one ethyl).

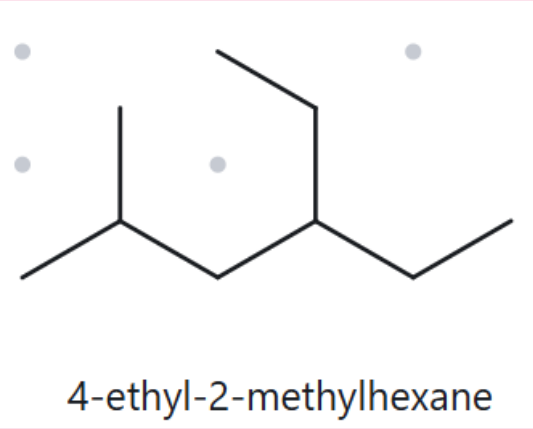
Numbering: left → right gives 2,4; right → left gives 3,5. First point of difference: 2 < 3, so number from the left.

Substituents: A→Z: ethyl on C4, methyl on C2 → 4-ethyl-2-methyl.

3 · MY NAME

4-ethyl-2-methylhexane

4 · THE ALKANE CHECK



Returned name:

4-ethyl-2-methylhexane

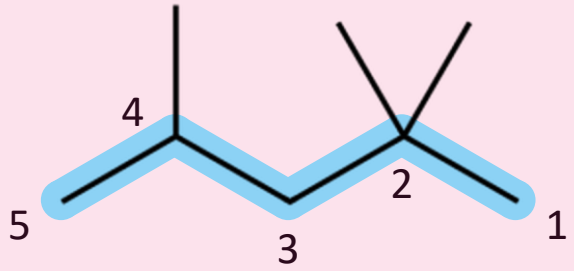
✓ **Match**

Same as my name (4-ethyl-2-methylhexane): parent, locants, substituents and punctuation all agree.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

Molecular formula: C_8H_{18}

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 5 C → pentane (the equal-length chains all carry three methyls).

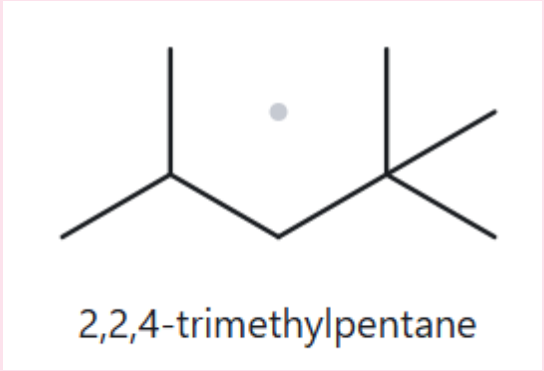
Numbering: right → left gives 2,2,4; left → right gives 2,4,4. First point of difference: 2 < 4, so number from the right.

Substituents: methyl on C2 (two) and on C4 → 2,2,4-trimethyl.

3 · MY NAME

2,2,4-trimethylpentane

4 · THE ALKANE CHECK



Returned name:

2,2,4-trimethylpentane

✓ **Match**

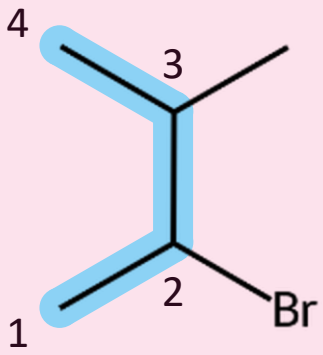
Same as my name (2,2,4-trimethylpentane): parent, locants, substituents and punctuation all agree.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A06 DEVELOPING Part A · Structure → Name

Molecular formula: $C_5H_{11}Br$

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 4 C → butane (bromine is a substituent).

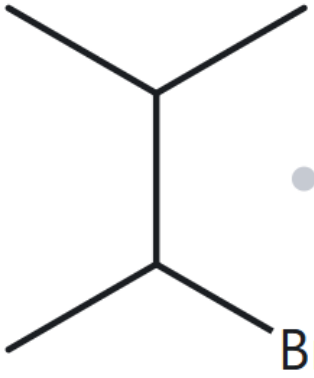
Numbering: bottom → top gives 2,3; top → bottom gives 2,3. Tie: bromo is first alphabetically, so it takes C2 (number from the bottom).

Substituents: A→Z: bromo on C2, methyl on C3 → 2-bromo-3-methyl.

3 · MY NAME

2-bromo-3-methylbutane

4 · THE ALKANE CHECK



2-bromo-3-methylbutane

Returned name:

2-bromo-3-methylbutane

✓ **Match**

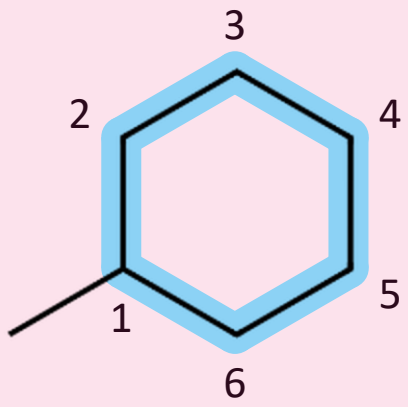
Same as my name (2-bromo-3-methylbutane): parent, locants, substituents and punctuation all agree.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A07 DEVELOPING Part A · Structure → Name

Molecular formula: C_7H_{14}

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: ring of 6 C → cyclohexane (the ring is the parent).

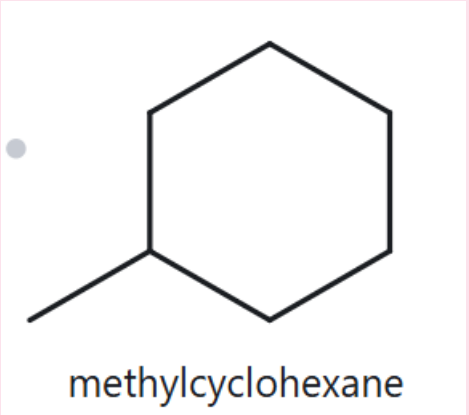
Numbering: one substituent: its carbon is C1; either way round gives the same name.

Substituents: methyl on C1 → methyl (no locant for a single substituent on a ring).

3 · MY NAME

methylcyclohexane

4 · THE ALKANE CHECK



Returned name:

methylcyclohexane

✓ **Match**

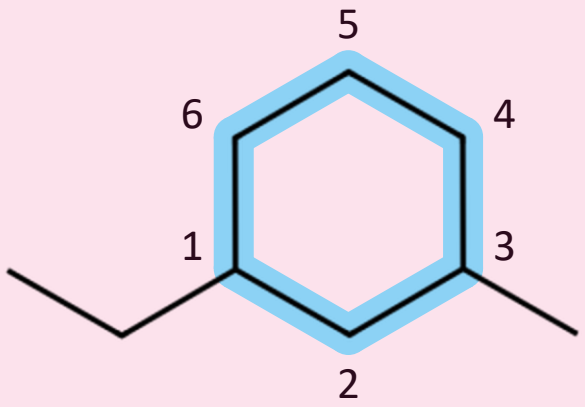
Same as my name (methylcyclohexane): parent, locants, substituents and punctuation all agree.

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A08 CHALLENGE Part A · Structure → Name

Molecular formula: C_9H_{18}

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: ring of 6 C → cyclohexane (the ring is the parent).

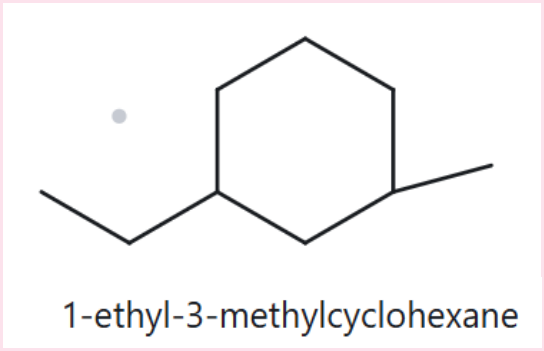
Numbering: short way round gives 1,3 from either substituent: tie. Ethyl is first alphabetically, so it takes C1.

Substituents: A→Z: ethyl on C1, methyl on C3 → 1-ethyl-3-methyl.

3 · MY NAME

1-ethyl-3-methylcyclohexane

4 · THE ALKANE CHECK



Returned name:

1-ethyl-3-methylcyclohexane

✓ **Match**

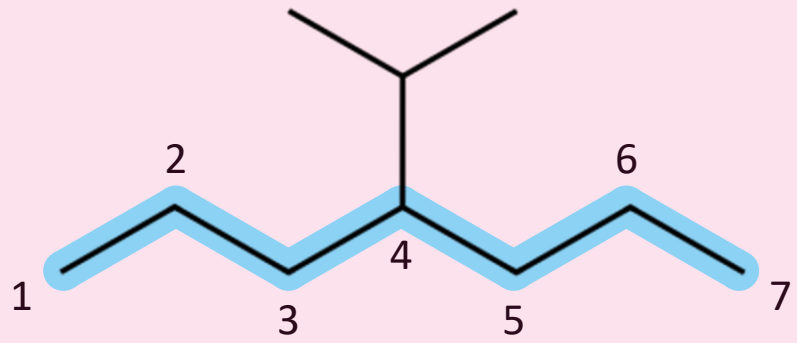
Same as my name (1-ethyl-3-methylcyclohexane): parent, locants, substituents and punctuation all agree.

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A09 CHALLENGE Part A · Structure → Name

Molecular formula: $C_{10}H_{22}$

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: longest chain = 7 C → heptane (left propyl arm + branch carbon + right propyl arm; through the branch only 6 C).

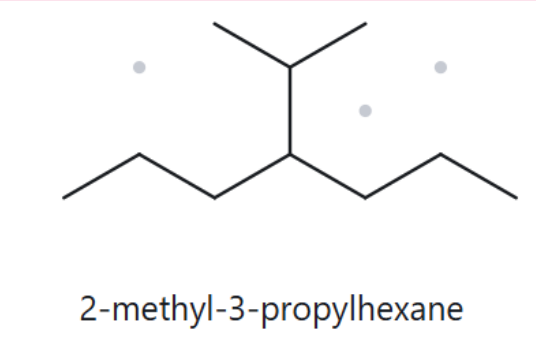
Numbering: left → right gives 4; right → left gives 4. Tie: both directions give the same name.

Substituents: isopropyl on C4 → 4-isopropyl.

3 · MY NAME

4-isopropylheptane

4 · THE ALKANE CHECK



Returned name:

2-methyl-3-propylhexane

✗ **Mismatch**

Correct name: **4-isopropylheptane** (= 4-(1-methylethyl)heptane).

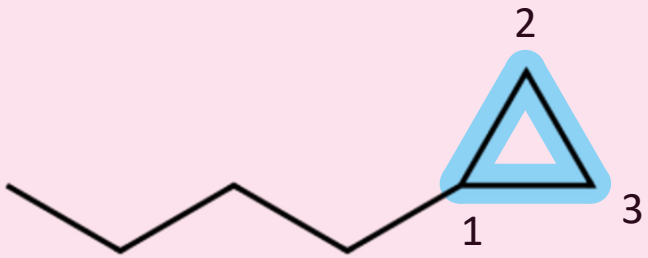
Rule not followed: the parent is the longest continuous chain. The tool used a 6-C chain (hexane) through the isopropyl branch; the longest chain is 7 C (heptane), through both propyl arms, so the branch is isopropyl on C4. Same compound ($C_{10}H_{22}$), but only 4-isopropylheptane obeys the rule.

✓ longest chain confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

A10 CHALLENGE Part A · Structure → Name

Molecular formula: C_7H_{14}

1 · NUMBERED PARENT



2 · NAMING TRAIL

Parent: ring of 3 C → cyclopropane (the ring is the parent, even though the chain has 4 C).

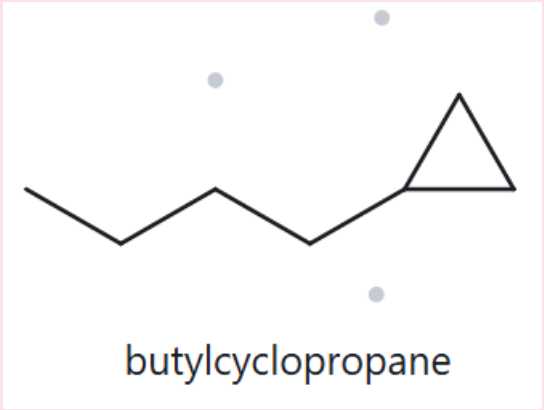
Numbering: one substituent: its carbon is C1; either way round gives the same name.

Substituents: butyl on C1 → butyl (no locant for a single substituent on a ring).

3 · MY NAME

butylcyclopropane

4 · THE ALKANE CHECK



Returned name:

butylcyclopropane

✓ **Match**

Same as my name (butylcyclopropane): parent, locants, substituents and punctuation all agree.

✓ longest chain/ring confirmed ✓ lowest locants ✓ alphabetical order ✓ punctuation ✓ formula counted

B01 FOUNDATION Part B · Name → Structure

Given name: 2-methylpropane

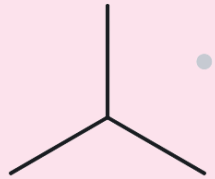
1 · NAME DECODED

Parent: propane → a 3-carbon chain.

Substituents: methyl on C2.

Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}.

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C_4H_{10}

Open chain: CnH2n+2 with n = 4 → C_4H_{10} ✓

4 · BACK-CHECK

Returned name: **2-methylpropane**

✓ Match — same as the given name

The tool returned 2-methylpropane: parent length, substituent identity and locants agree with the name I decoded, and the formula agrees.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B02 FOUNDATION Part B · Name → Structure

Given name: **3-ethylpentane**

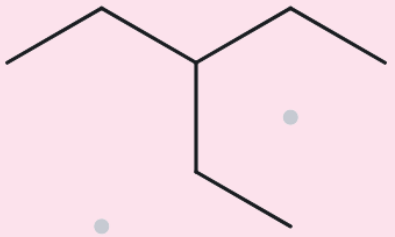
1 · NAME DECODED

Parent: pentane → a 5-carbon chain.

Substituents: ethyl on C3 (the middle carbon).

Expected formula: 5 + 2 = 7 C; open chain → C_7H_{16}.

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: **C_7H_{16}**

Open chain: CnH2n+2 with n = 7 → C_7H_{16} ✓

4 · BACK-CHECK

Returned name: **3-ethylpentane**

✓ Match — same as the given name

The tool returned 3-ethylpentane: parent length, substituent identity and locants agree with the name I decoded, and the formula agrees.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B03 FOUNDATION Part B · Name → Structure

Given name: 2,2-dimethylbutane

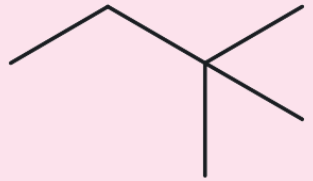
1 · NAME DECODED

Parent: butane → a 4-carbon chain.

Substituents: two methyl groups, both on C2.

Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}.

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C_6H_{14}

Open chain: CnH2n+2 with n = 6 → C_6H_{14} ✓

4 · BACK-CHECK

Returned name: 2,2-dimethylbutane

✓ Match — same as the given name

The tool returned 2,2-dimethylbutane: parent length, substituent identity and locants agree with the name I decoded, and the formula agrees.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B04 DEVELOPING Part B · Name → Structure

Given name: **3-ethyl-2-methylpentane**

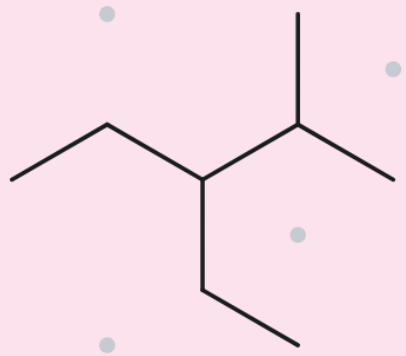
1 · NAME DECODED

Parent: pentane → a 5-carbon chain.

Substituents: ethyl on C3; methyl on C2 (neighbouring carbons).

Expected formula: 5 + 2 + 1 = 8 C; open chain → C_8H_{18}.

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: **C_8H_{18}**

Open chain: CnH2n+2 with n = 8 → C_8H_{18} ✓

4 · BACK-CHECK

Returned name: **3-ethyl-2-methylpentane**

✓ Match — same as the given name

The tool returned 3-ethyl-2-methylpentane: parent length, substituent identity and locants agree with the name I decoded, and the formula agrees.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B05 DEVELOPING Part B · Name → Structure

Given name: **1-chloro-2-methylpropane**

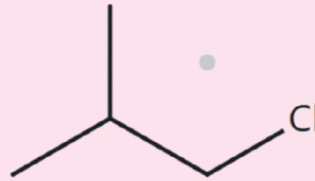
1 · NAME DECODED

Parent: propane → a 3-carbon chain.

Substituents: chloro on C1 (terminal carbon); methyl on C2.

Expected formula: 3 + 1 = 4 C; open chain → C_4H_{10}; Cl replaces one H → C_4H_9Cl.

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: **C_4H_9Cl**

Open chain: CnH2n+2 with n = 4 → C_4H_{10}; Cl replaces one H → C_4H_9Cl ✓

4 · BACK-CHECK

Returned name: **1-chloro-2-methylpropane**

✓ Match — same as the given name

The tool returned 1-chloro-2-methylpropane: parent length, substituent identity and locants agree with the name I decoded, and the formula agrees.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B06 DEVELOPING Part B · Name → Structure

Given name: **1,2-dimethylcyclopentane**

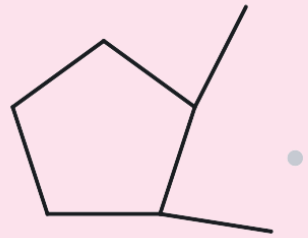
1 · NAME DECODED

Parent: cyclopentane → a 5-carbon ring.

Substituents: methyl on C1 and on C2 (adjacent ring carbons).

Expected formula: 5 + 2 = 7 C; one ring → C_7H_{14}.

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: **C_7H_{14}**

One ring: CnH2n with n = 7 → C_7H_{14} ✓

4 · BACK-CHECK

Returned name: **1,2-dimethylcyclopentane**

✓ Match — same as the given name

The tool returned 1,2-dimethylcyclopentane: parent length, substituent identity and locants agree with the name I decoded, and the formula agrees.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B07 DEVELOPING Part B · Name → Structure

Given name: **4-ethyl-3,3-dimethylheptane**

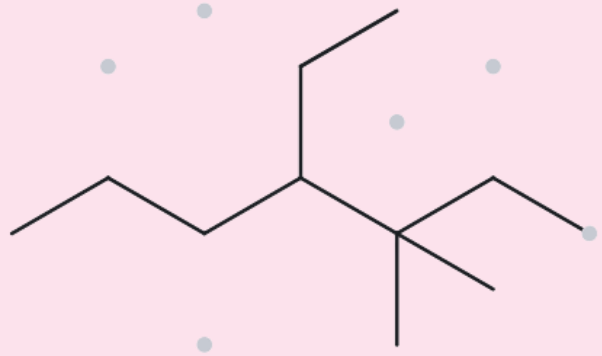
1 · NAME DECODED

Parent: heptane → a 7-carbon chain.

Substituents: ethyl on C4; two methyl groups, both on C3.

Expected formula: 7 + 2 + 2 = 11 C; open chain → $C_{11}H_{24}$.

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: **$C_{11}H_{24}$**

Open chain: CnH2n+2 with n = 11 → $C_{11}H_{24}$ ✓

4 · BACK-CHECK

Returned name: **4-ethyl-3,3-dimethylheptane**

✓ Match — same as the given name

The tool returned 4-ethyl-3,3-dimethylheptane: parent length, substituent identity and locants agree with the name I decoded, and the formula agrees.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B08 CHALLENGE Part B · Name → Structure

Given name: tert-butylcyclohexane

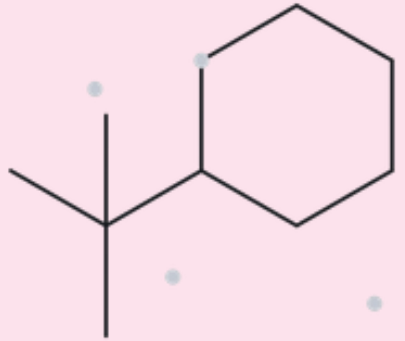
1 · NAME DECODED

Parent: cyclohexane → a 6-carbon ring.

Substituents: tert-butyl on C1 (locant omitted: one substituent). tert-butyl = $(CH_3)_3C–$ = (1,1-dimethylethyl) = 2-methylpropan-2-yl (App. B.3).

Expected formula: 6 + 4 = 10 C; one ring → $C_{10}H_{20}$.

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: $C_{10}H_{20}$

One ring: CnH2n with n = 10 → $C_{10}H_{20}$ ✓

4 · BACK-CHECK

Returned name: (1,1-dimethylethyl)cyclohexane

✗ Mismatch in wording only — same compound ✓

Why: The Alkane writes tert-butyl in its bracketed systematic form, (1,1-dimethylethyl) (App. B.3, C.6). Both names describe the same group, $(CH_3)_3C–$, on a 6-C ring: $C_{10}H_{20}$ either way. The drawing is correct, so it is not changed; the given name tert-butylcyclohexane is an accepted equivalent and earns full credit.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B09 CHALLENGE Part B · Name → Structure

Given name: **3-ethyl-1,1-dimethylcyclohexane**

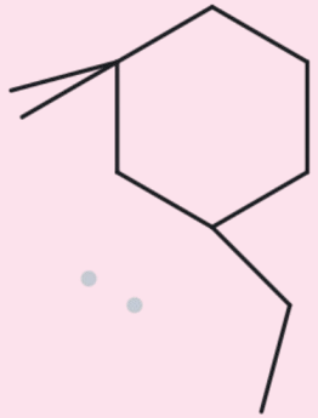
1 · NAME DECODED

Parent: cyclohexane → a 6-carbon ring.

Substituents: two methyl groups on C1 (same carbon); ethyl on C3 (C2 lies between them).

Expected formula: 6 + 2 + 2 = 10 C; one ring → $C_{10}H_{20}$.

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: **$C_{10}H_{20}$**

One ring: CnH2n with n = 10 → $C_{10}H_{20}$ ✓

4 · BACK-CHECK

Returned name: **3-ethyl-1,1-dimethylcyclohexane**

✓ Match — same as the given name

The tool returned 3-ethyl-1,1-dimethylcyclohexane: parent length, substituent identity and locants agree with the name I decoded, and the formula agrees.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded

B10 CHALLENGE Part B · Name → Structure

Given name: 2-ethylbutane

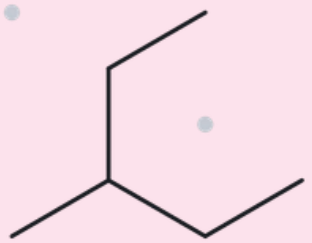
1 · NAME DECODED

Parent: butane (as printed) → a 4-carbon chain.

Substituents: ethyl on C2 — drawn exactly as the name says.

Expected formula: 4 + 2 = 6 C; open chain → C_6H_{14}.

2 · BOND-LINE STRUCTURE (THE ALKANE)



3 · FORMULA CHECK

Counted from the drawing: C_6H_{14}

Open chain: CnH2n+2 with n = 6 → C_6H_{14} ✓

4 · BACK-CHECK

Returned name: 3-methylpentane

✗ Mismatch — the given name is wrong, not the drawing

Correct name: 3-methylpentane (given: 2-ethylbutane)

Rule broken: the parent must be the longest continuous chain. In 2-ethylbutane the ethyl group lengthens the chain: the longest path is 5 C (pentane), running through the ethyl. What is left is one methyl on the middle carbon, C3. The drawing is exactly what the name says (C_6H_{14}), and the structure itself gives the correct name.

✓ parent length matches ✓ substituent positions match ✓ formula agrees ✓ back-check recorded